Final Report for the project:

Analysis and Visualization of Network Attacks using an IDS Dataset

Edited by:

Martyna Kochalska

Piotr Szkoda

Marten Tschatschanidse

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Professor: Ladji Adiaviakoye

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1. Introduction

Cybersecurity represents a critical discipline dedicated to protecting digital assets, information systems, and network infrastructure from malicious activities and unauthorized access in our increasingly interconnected world [1]. As organizations and individuals become more dependent on digital technologies and internet connectivity, the attack surface for malicious actors has expanded exponentially, creating vulnerabilities that span from individual devices to complex enterprise networks and critical infrastructure systems [2]. Network attacks, which encompass a broad spectrum of malicious activities including unauthorized intrusions, data breaches, denial-of-service attacks, malware propagation, and advanced persistent threats, pose significant risks to data integrity, system availability, and organizational operations [3]. These attacks exploit various vulnerabilities in network protocols, system configurations, and human behaviors, often resulting in substantial financial losses, privacy violations, and disruption of essential services [4]. To counter these evolving threats, Intrusion Detection Systems (IDS) have emerged as fundamental components of cybersecurity infrastructure, designed to monitor network traffic and system activities in real-time to identify and alert security personnel about suspicious or malicious behaviors [5]. IDS technologies operate by analyzing network packets, system logs, and behavioral patterns against known attack signatures and anomaly detection algorithms, serving as an early warning system that enables organizations to detect, investigate, and respond to security incidents before they can cause significant damage [6]. The continuous evolution of both attack methodologies and defensive technologies has created a dynamic cybersecurity landscape where the effectiveness of protection mechanisms depends heavily on the ability to rapidly detect, analyze, and visualize complex attack patterns within vast amounts of network data [7].

This project addresses the critical need for effective analysis and visualization of network attacks through the systematic examination of IDS datasets. By combining advanced data analysis techniques with innovative visualization approaches, this work aims to contribute to the broader understanding of network attack patterns and improve the effectiveness of cybersecurity operations. The integration of data analysis with interactive visualization tools provides a framework for both automated detection and human-interpretable analysis of network security data, addressing key challenges in contemporary cybersecurity practice.

2. Dataset

The CIC-IDS 2017 dataset, created by the Canadian Institute for Cybersecurity, is one of the most comprehensive and realistic publicly available datasets for intrusion detection research. It aims to simulate a real-world enterprise network environment by capturing both normal user activities and a wide variety of contemporary cyberattack scenarios. The dataset was collected over a period of five days in a controlled network setup involving more than 25 users, producing network traffic that closely resembles today's corporate network behavior.

A key advantage of CIC-IDS 2017 is the precise labeling of each network flow as either benign or belonging to a specific attack category, which facilitates supervised machine learning approaches. Each network flow in the dataset is richly annotated with detailed features, including timestamps, source and destination IP addresses and ports, protocol specifics, packet and byte counts, flow duration, and various application-layer attributes. This allows cybersecurity researchers and data scientists to build, train, and benchmark detection models against realistic attack behavior. The dataset's inclusion of diverse attack types and modern network architecture makes it highly relevant for developing detection mechanisms capable of recognizing both known and emerging threats.

Due to its size, realism, thorough annotation, and variety, CIC-IDS 2017 has become a standard benchmark dataset in cybersecurity research for testing and comparing intrusion detection systems, helping to improve network security through data-driven methods [8, 9].

The CIC-IDS 2017 dataset encompasses a wide variety of network attack types, reflecting many of the common cybersecurity threats faced in real-world environments. This includes distributed denial-of-service (DDoS) attacks, brute force attempts targeting FTP and SSH services, infiltration efforts, web application exploits such as SQL injection and cross-site scripting (XSS), and advanced threats like exploitation of the Heartbleed vulnerability.

Denial-of-Service (DoS) attacks within the dataset, including Hulk, GoldenEye, Slowloris, Slowhttptest and DDoS focus on overwhelming network resources through different flooding techniques, often causing service interruptions. The dataset also covers reconnaissance techniques like Portscan, which identify vulnerable network services, alongside other intrusion methods such as botnet activity and systematic credential guessing through brute-force attacks.

The following table summarizes the primary attack types in the dataset, highlighting their key characteristics and the recommended way of detection [10–13].

Attack Type	Short Description
DoS Hulk	Floods web servers with randomized HTTP requests to exhaust resources and deny service.
DoS GoldenEye	Sends slow or incomplete HTTP requests, keeping connections open and causing server thread exhaustion.
DoS Slowloris	Maintains many open connections with partial HTTP requests, exhausting the server's capacity.
DoS Slowhttptest	Uses slow, incomplete HTTP requests to fill server connections and disrupt normal service.
DDoS	Uses multiple devices to flood a target with traffic, overwhelming bandwidth and resources.
Portscan	Scans network ports to identify open services and vulnerabilities.
FTP-Patator	Performs brute-force attacks on FTP servers by rapidly guessing login credentials.
SSH-Patator	Brute-force attack targeting SSH logins with automated password guessing.
Bot	Compromised devices operated remotely, used for coordinated attacks or malware spread.
Web Attack Brute Force	Attempts to gain access by systematically guessing web application credentials.
Web Attack XSS	Injects malicious scripts into web pages to steal data or manipulate users.
Web Attack SQL Injection	Inserts harmful SQL queries via input fields to access or corrupt backend data.
Infiltration	Gains unauthorized access to internal network resources, often for data exfiltration.
Heartbleed	Exploits OpenSSL vulnerability to read sensitive server memory and expose data.

Table 1: Attack types and short description

3. Dataset Acquisition, Preparation, and Visualization

We started the project by downloading an initial IDS dataset from Kaggle. However, after some preliminary investigation, we realized that this publicly shared version lacked critical information, such as timestamps, which limited our ability to conduct a thorough data analysis. Therefore, we decided to request the official dataset from the University of New Brunswick, which was considerably more complete and better suited for detailed data exploration. This dataset provided us with comprehensive network traffic logs, including a wide range of features essential for our analysis and visualization tasks.

Once we had acquired the full dataset, we focused on data cleaning and preparation. Given the dataset's size of approximately 1.54 GB after expansion and the presence of numerous features, many of which were less relevant or redundant, we began by removing missing values since they constituted only a minor portion and did not significantly affect the analysis. We then developed scripts to thoroughly describe and understand the dataset structure and the significance of various features. To optimize our workload and improve the quality of insights, we created filtered versions of the dataset by reducing the amount of benign, or normal user, traffic and by selecting specific time periods when certain attacks surpassed a threshold number of samples. This careful sampling method ensured that the reduced datasets remained representative of the overall population, preserving the essential characteristics of network traffic and attack patterns.

When it came to visualization, we initially contemplated using Streamlit due to our prior experience with it and its simplicity for quick prototyping. However, we recognized that Streamlit might struggle with the volume of data we intended to handle. Consequently, we shifted our focus to Dash, a more professional and scalable framework widely used for creating interactive dashboards. We explored Plotly, which integrates seamlessly with Dash, to generate interactive, clean charts that could capture detailed analyses for various attack types. But we encountered performance issues: Plotly’s plots became resource-heavy and slow as dataset size increased, posing a significant challenge.

To address these performance constraints and given that our team had only three members instead of four, we decided to revise and simplify our dashboard design. We consolidated the numerous attack types into four primary groups, reducing the number of charts and categories. Through random sampling of data from each traffic category, we were able to significantly reduce the dataset size without compromising key statistical averages. This approach allowed

us to generate optimized plots capable of loading almost in real-time, resulting in a dashboard that, while simpler and less flashy, was interactive, responsive, and functional.

Alongside the dashboard, for more detailed and accurate analyses, we crafted additional plots in Matplotlib using the entire dataset. These plots provided deeper statistical insights and were included in the final report and presentation to complement the dashboard's interactive visualizations.

In summary, our project implementation followed a structured and pragmatic approach. Starting from obtaining the most suitable dataset, through rigorous cleaning and intelligent data reduction, to building an effective interactive dashboard balancing performance and functionality. The combination of Dash and Plotly enabled us to present an insightful visualization of IDS network attacks, while the supplementary Matplotlib charts enriched the overall analysis and reporting.

4. Results

The first figure (Figure 1), titled "Distribution of All Attack Types," presents a comprehensive overview of the frequency of various attack types recorded in the dataset. The bar chart depicts the number of network flows associated with each attack category, clearly illustrating the scale and prevalence of different intrusion attempts.

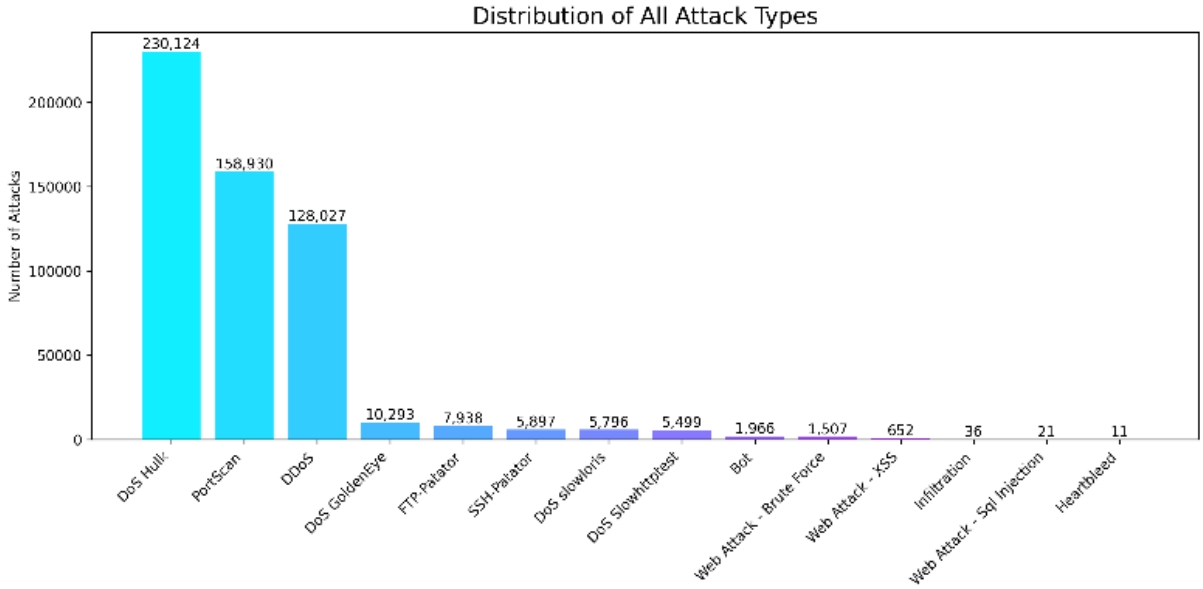

Figure 1: Distribution of all attack types

From the chart, it is evident that DoS Hulk attack dominates the dataset with 230,124 recorded instances, making it the most frequent attack type by a significant margin. This high count underscores the pervasive nature of Denial-of-Service (DoS) attacks, which aim to overwhelm network resources and disrupt service availability.

The most prominent following attack types are PortScan (158,930 instances) and DDoS attacks (128,027 instances), demonstrating that automated reconnaissance and distributed denial-of-service attempts constitute the bulk of malicious activity in this dataset.

All other attack types including DoS GoldenEye, FTP-Patator, SSH-Patator, DoS Slowloris, DoS Slowhttptest, Bot, Web Attack Brute Force, Web Attack XSS, Infiltration, Web Attack SQL Injection, and Heartbleed are far less frequent, each with under 11,000 observed events. These collectively represent a minor proportion of the total attacks detected, suggesting that large-scale volumetric and probing activities are most common in the captured network traffic.

The next figure (Figure 2) records the exact timing of each attack for every day of the simulated working week, with five panels corresponding to Monday through Friday. The horizontal axis

focuses on selected time periods to enhance readability, while the vertical axis lists distinct attack types.

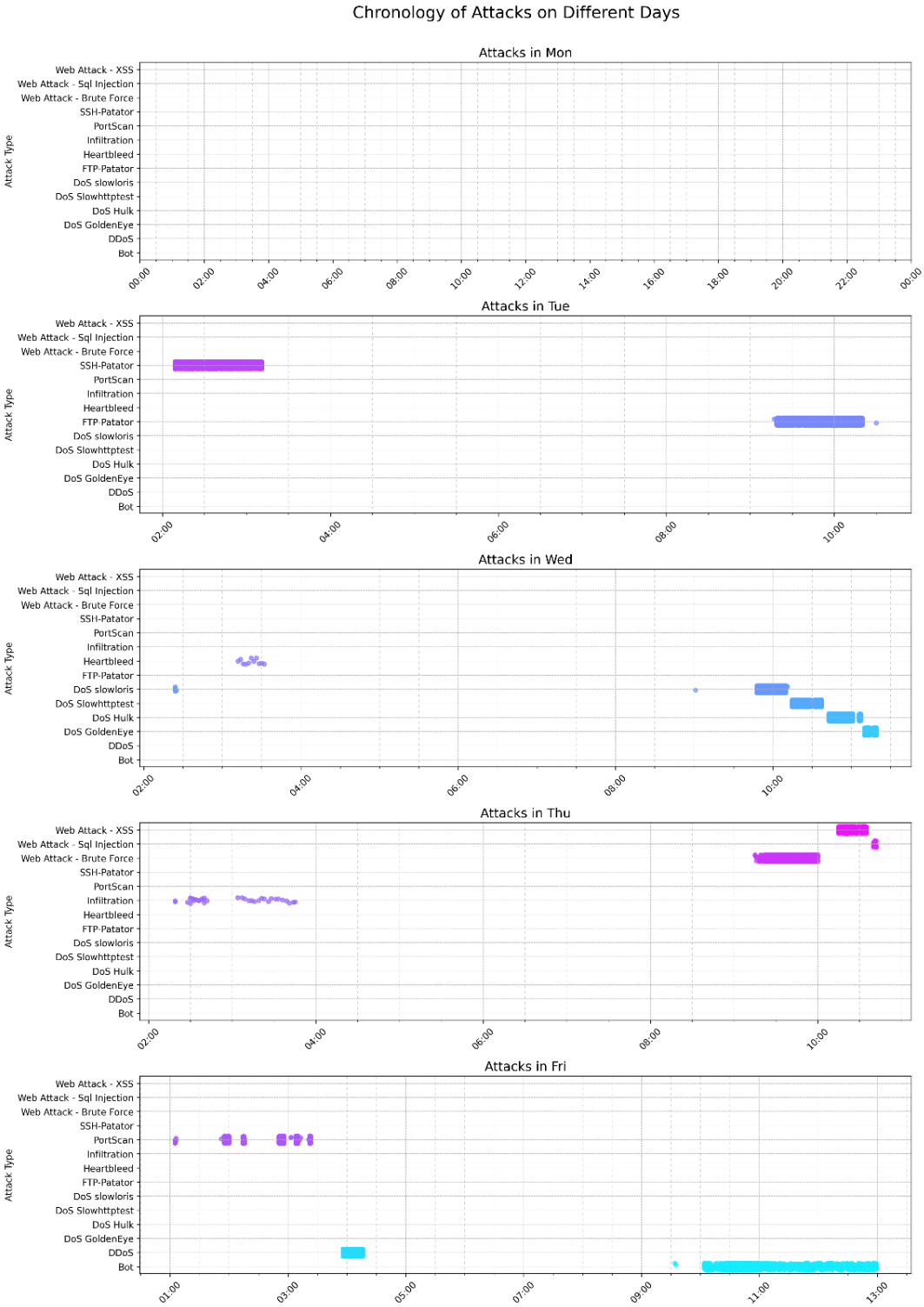

Figure 2: Chronology of attacks on different days

This chronological overview reveals several key insights:

- Certain attacks, notably denial-of-service (DoS), cluster tightly within narrow time windows, reflecting their intent to overwhelm servers through concentrated traffic bursts.
- Other attacks are more sparsely and irregularly distributed, suggesting deliberate randomization, potentially to evade detection or due to manual, human-driven execution.
- Most attack types appear only once during the week, indicating that much of the dataset consists of singular intrusion events. This characteristic must be considered during analysis, as it may limit the extraction of a definitive behavioral fingerprint for each specific attack category.

Figure 3 presents four box plots, each depicting statistical distributions for different network metrics across attack groups namely, packets per second, bytes per flow, flow duration, and bytes per packet.

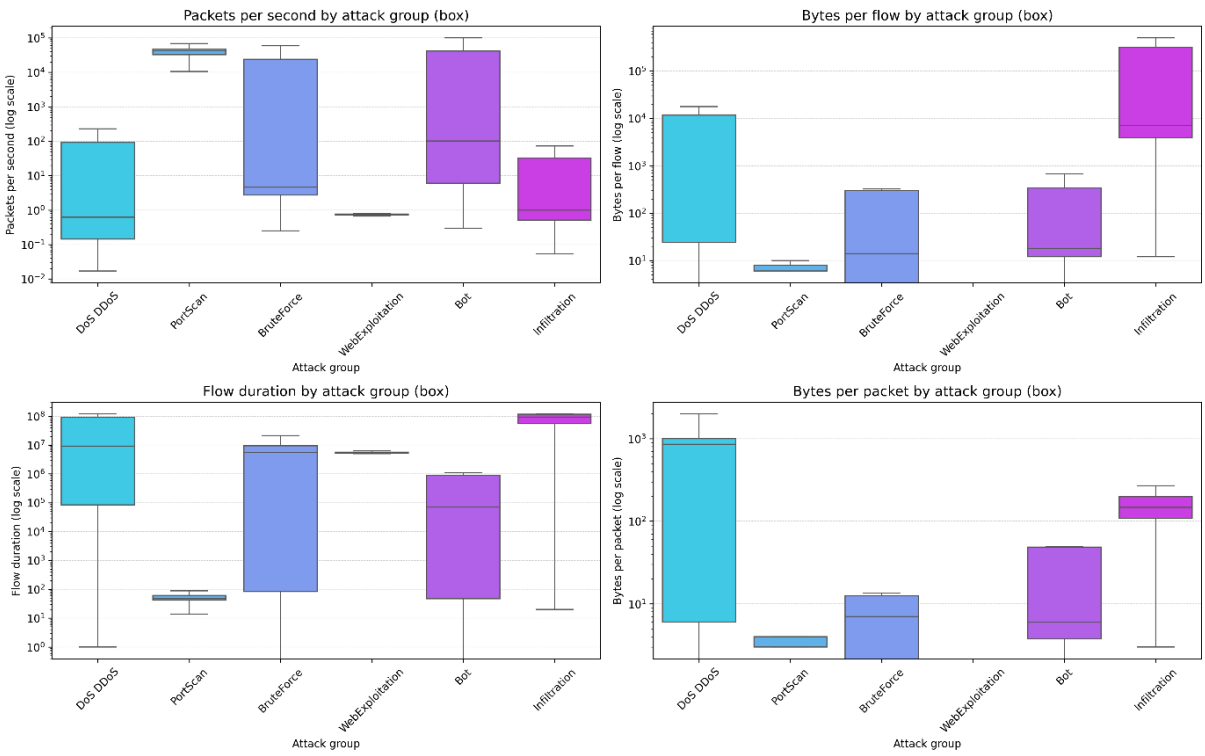

Figure 3: Boxplot to describe the range of data distribution

Interpreting the box leads to the following conclusions:

- Port Scan attacks generate a high number of very short, minimal flows, consistent with their reconnaissance nature of quickly probing available ports.
- Web exploitation is barely visible, reflecting a lack of sufficient data for meaningful analysis of this category.
- The DoS group shows relatively low packets per second but high flow duration, attributed to the inclusion of Slowloris and Slowhttptest attacks, which deliberately create prolonged connections that exhaust server resources; this contrasts with other faster DoS methods and is further substantiated in their dedicated analysis.
- Brute Force and Bot attacks exhibit high packet rates but low bytes per flow, indicating frequent but simple requests—often originating from basic automation scripts or services.
- The Infiltration attack group is characterized by extremely long flow duration, suggesting the attacker may have gained lasting access to a server, potentially maintaining a stable session for ongoing exploration or data exfiltration. This warrants close scrutiny, especially when such flows originate from unfamiliar sources outside the organization.

This heatmap visualization (Figure 4) highlights which attack types most frequently targeted specific destination ports. Port Scan attacks, as expected, dominate nearly all port categories by probing a wide range of services. Bruteforce attacks (Patator) primarily focus on SSH (port 22) and FTP (port 21), reflecting their attempt to gain unauthorized access via these protocols. Bot and DoS/DDoS attacks concentrate on HTTP-related ports to blend in with typical traffic patterns, making detection more difficult. An interesting observation for infiltration attacks is an increased presence on ports like 4444 and a slight rise at port 444, which are not standard and may indicate usage of backdoor or reverse shell techniques for remote access. This underlines the importance of monitoring rarely used ports, as unusual activity there can reveal stealthy intrusion methods.

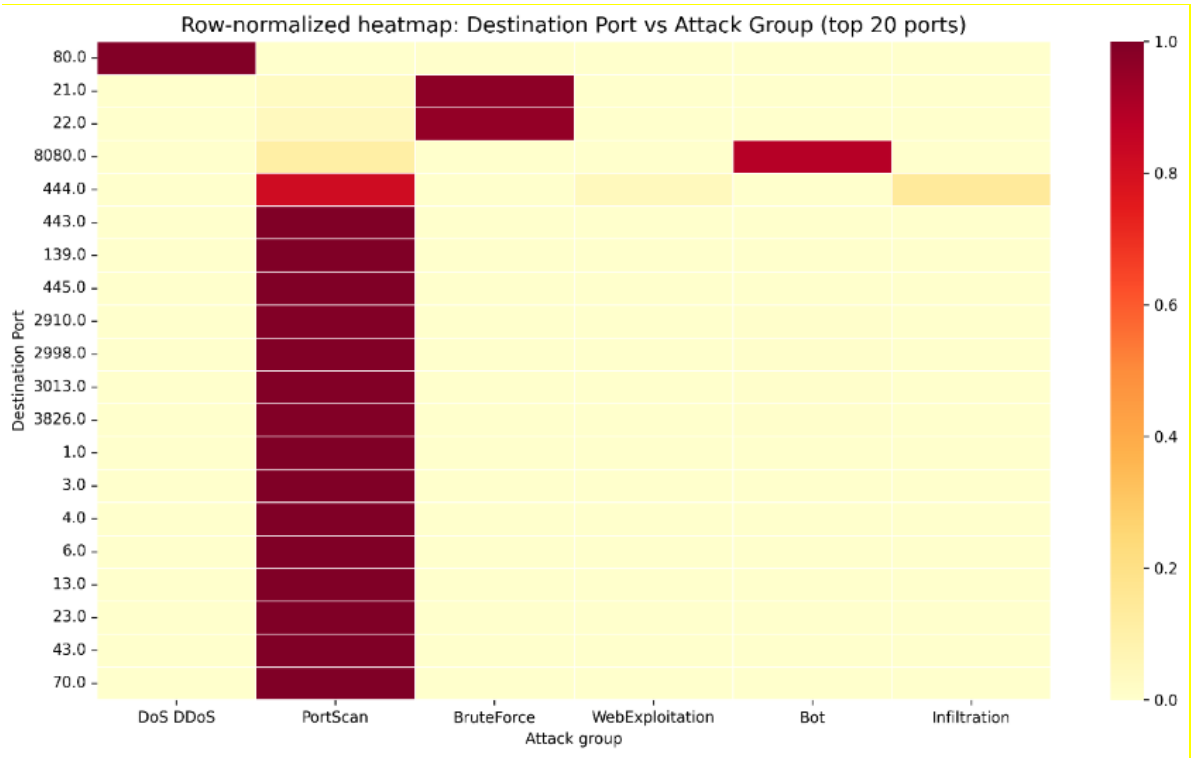

Figure 4: Row-normalized Heatmap: Destination Port vs Attack Group

The heatmap in Figure 5 presents binary Spearman correlations between network features and attack types, filtered for statistical significance. Each cell indicates whether a specific feature consistently deviates either above or below from normal user (benign) traffic when a given attack occurs, using only those combinations with strong patterns (p-value < 0.05).

The color scale ranges from blue (negative values, indicating the feature is consistently lower than in benign flows) to red (positive, consistently higher), giving a visual summary of how each attack type alters network behavior compared to normal usage. The hierarchical clustering on both axes connects attacks and features that show similar correlation profiles, making it easier to spot related patterns.

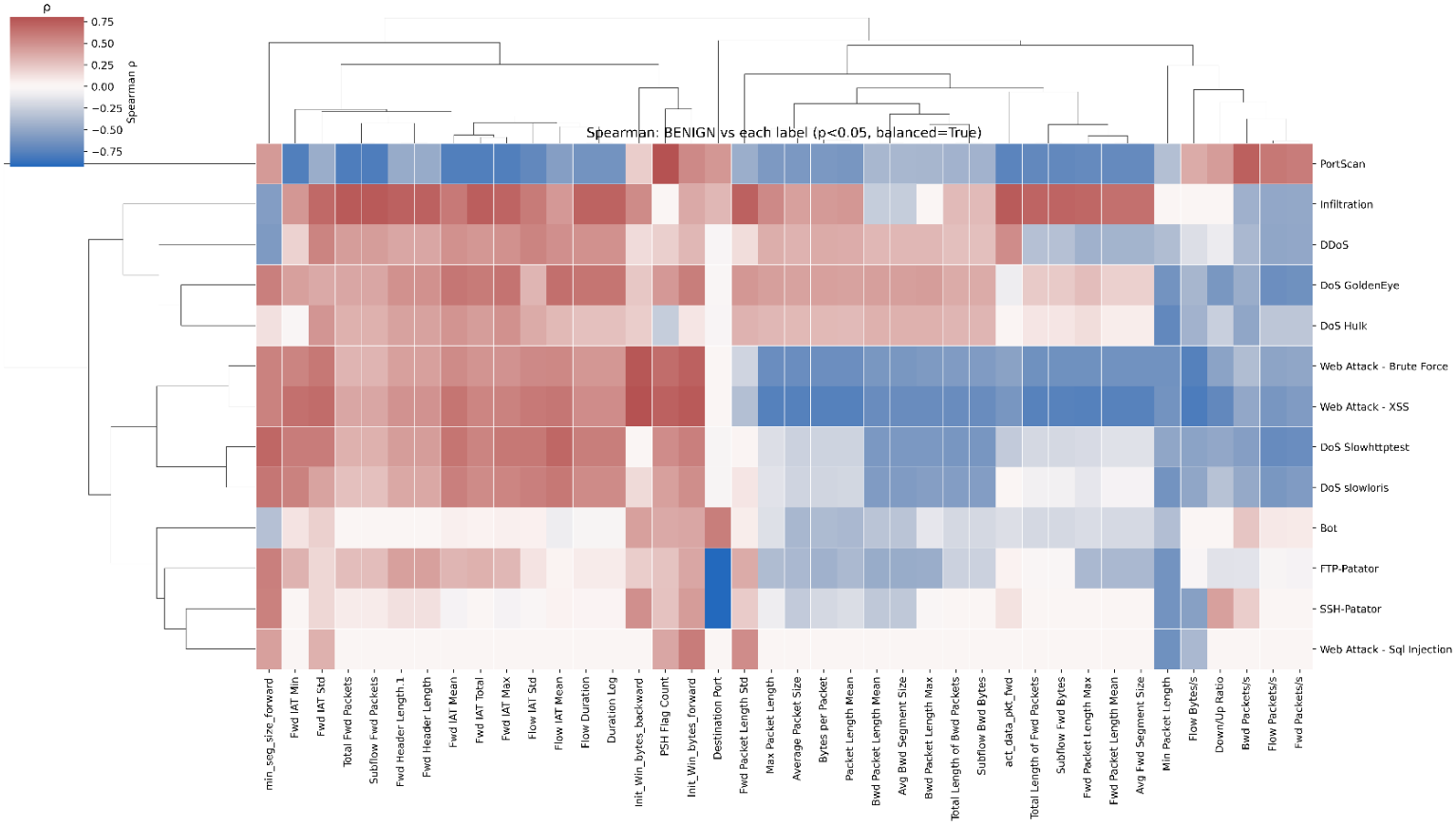

Figure 5: Spearman BENIGN vs. each label

Key conclusions visible from the plot:

- Caution is essential with attack types that have less than a few hundred instances, as their characteristics could be skewed by outliers or specific attacker behaviors.
- DoS Slowhttptest and Slowloris attacks are distinct from DDoS, DoS Hulk, and GoldenEye; the former are characterized by persistently low packet and segment lengths, focusing on maintaining server connections rather than overwhelming bandwidth.
- PortScan is unique among attacks, showing very low packet lengths and inter-arrival times but generating a large number of flows. A higher PSH flag count suggests the scan sometimes operates at the application layer.
- Patator (brute force) attacks often target specific ports and are capable of blending in closely with normal network traffic patterns.

- Infiltration attacks are notable for an increased count of active forward-directed packets, which can result from compromised devices sending data out or sustained reverse TCP tunnels for continuous remote access

This box plot Figure 6 visualizes the flow duration for different attack types and benign traffic.

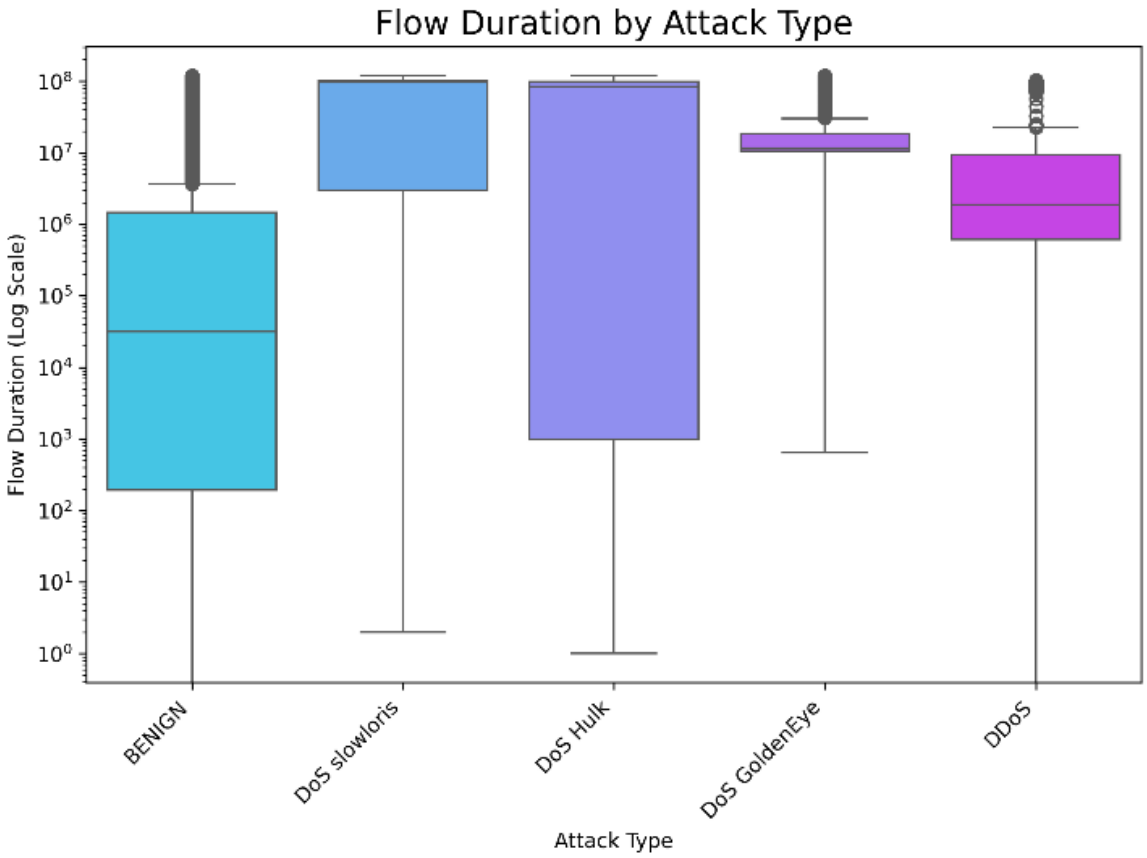

Figure 6: Flow Duration by attack type

Notably, DoS Slowloris stands out with a dramatically high median flow duration, which aligns perfectly with its attack method of maintaining long-lived connections to exhaust server resources. DoS GoldenEye also shows a high flow duration, possibly mimicking legitimate peak-time traffic patterns on the server. DDoS, DoS Hulk, and other grouped attacks generally have shorter durations, but the variance is considerable, reflecting the diversity of real-world implementations, not all strictly follow their textbook pattern.

Benign traffic, meanwhile, displays a significant range of outliers, including some flows with exceptionally high durations. This overlap means it is not possible to simply set a precise threshold (for example, above 10^7 microseconds) to reliably distinguish DoS/DDoS attacks from normal activity. Overall, the box plot demonstrates that while certain DoS attacks can be

characterized by their prolonged flows, distinguishing them in practice requires considering distribution as a whole rather than relying on a fixed cutoff.

This scan trace plot (Figure 7) illustrates how, within a brief period, the PortScan attack targets nearly all available ports in successive waves.

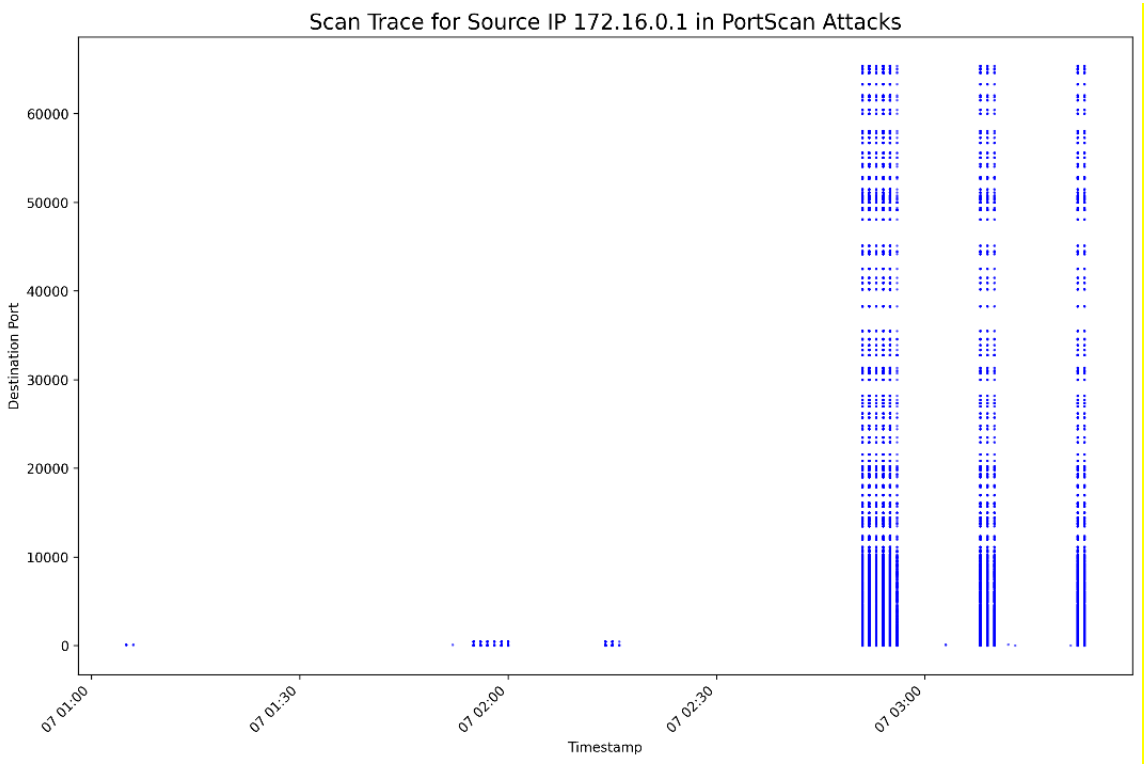

Figure 7: Scan Trace for Source IP 172.16.0.1 in PortScan Attacks

The pattern is highly consistent and does not attempt to conceal itself by spreading activity over time; in this particular example, the scan is performed very openly. Notably, the scan focuses primarily on ports below 100,000, aligning with the practice of probing commonly used ports.

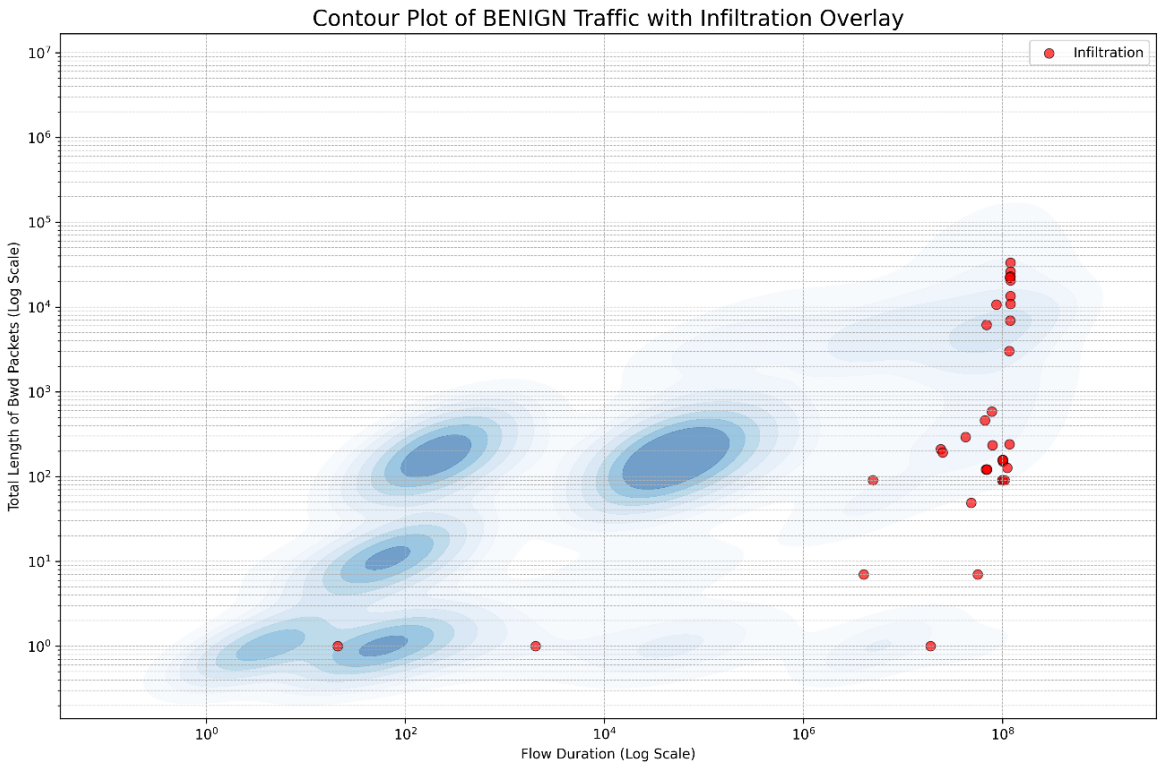

Figure 8: Contour Plot of BENIGN Traffic with infiltration overlay

The contour plot (Figure 8) allows a direct comparison between normal user traffic (displayed as blue clusters) and infiltration events (shown as red dots), highlighting how these singular attacks differ within the broader context of typical flows. Infiltration attacks stand out with extraordinarily long flow durations, yet their traffic often blends with lightly used legitimate patterns, such as SSH activity, and represents only a small fraction of all user traffic. While the length of backward packets is sometimes elevated during infiltration, it is less distinct than flow duration. Flow duration emerges as the most reliable red flag for identifying infiltration attacks and should be the first point of investigation, alongside a review of protocols and suspicious IP addresses behind anomalous sessions: for example, distinguishing between an authorized administrator working remotely and a potential intruder maintaining a prolonged SSH connection from outside the organization.

5. Summary

In summary, the analysis of the IDS dataset provides comprehensive insights into the behavior and characteristics of various network attack types. Our visualizations demonstrate that volumetric and reconnaissance-based attacks such as DoS Hulk, PortScan, and DDoS dominate the dataset, highlighting the persistent threat posed by automated and large-scale intrusion attempts. Brute force and bot attacks, while less frequent, reveal distinctive high-rate but simple activity patterns, emphasizing the diversity of methods employed by attackers.

Temporal analysis confirmed that many attacks are condensed within short time windows, consistent with the disruptive strategy of overwhelming victims, while other, sparser attack occurrences suggest randomized or even manually controlled intrusions. The varied distribution and correlation of key network features for each attack type, revealed through box plots and heatmaps, enabled a nuanced differentiation of attack behaviors uncovering, for instance, the prolonged duration of infiltration flows or the rapid probing nature of port scans.

Altogether, these findings demonstrate the complexity of monitoring modern networks in the face of evolving cyber threats. By combining careful data preparation, detailed statistical analysis, and targeted visualization, we have established a framework for recognizing and understanding anomalous patterns that may indicate emerging security incidents. This not only aids in threat detection but also lays the foundation for more effective and adaptive defense strategies in future security operations.

Out of the gathered information, a detailed outline of the different attacks, the recommended of detection and additional information is summarised in Table 2.

The complete interactive dashboard is available under the following link: https://ids-analyzer-dashboard.onrender.com/.

Overall, the dataset is valuable for gaining an overview and conducting descriptive analyses of different attack behaviors. However, caution should be exercised when generalizing findings to real-world scenarios, as the data may not capture the full variability of live network traffic. For machine learning applications, relying on this dataset alone may lead to overfitting due to limited variation and the presence of isolated flow events. Therefore, complementing it with additional IDS data sources is recommended to enhance robustness and generalizability in automated detection tasks.

Attack Name	Recommended way of detection	Additional information
DDoS	sudden spike in packets per second, usually very high source count abnormal to usual traffic	Attacks are very dense to overwhelm servers.
DoS Hulk	monitor for long flow duration and low packets per second	Surprisingly, in our case detected flow duration was very long, which is not intuitive with the way this attack is usually designed.
DoS GoldenEye	quick open/close of many HTTP connections	This attack tends to mimic normal traffic peaks and outliers making it hard to distinct.
DoS Slowloris	drastically high flow duration, consistently lower packet per segment length	This slowly chokes the server, does not work in bursts like many other DoS attacks
DoS Slowhttptest	numerous long lived HTTP connections, lower packet and segment length	Similar to slow-loris, aims to plug the server.
FTP-Patator	repeated TCP connections from singular IP to port 21 with USER login attempts	In our case we detected high packet rates with low flow size (around 20 bytes).
SSH-Patator	Repeated connections to port 22 with failed authentications	Like FTP-Patator it maintains high packet rate with relatively low flow size.
Web Attack (Brute Force, SQL Inj., XSS)	requests containing SQL or script tokens, distinct destination port usage	Web exploitation is barely visible in our plots, that is cause by insufficient and very small available data sample.
Heartbleed	TLS requests with declared payload length larger than actual payload length	Specific and very old vulnerability that should rather be patched than just detected.
Port Scan	large number of connections to a large range of ports possibly from one source, high PSH flag count	In our scenario it occurs in consistent “waves” hitting almost all ports at once.
Bot	high packet rate with very low flow size, uniform time intervals	Tends to stay on HTTP ports blending with usual user traffic.
Infiltration	extraordinary high flow duration, high number of active packets sent forward, use of protocols reserved for administrative purposes	In our scenario the attacker is abusing port 4444, it is not a dead-giveaway but many tools abuse usually the same ports if not re-configured.

Table 2: Attack types, recommended ways of detection and additional information

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